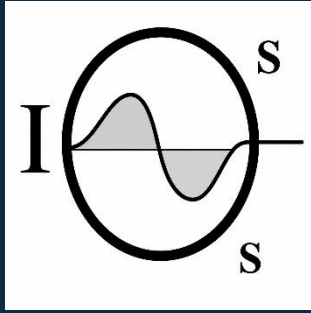




INSPECTION & SPECIFICATION SERVICES CC.

CONTROLLED ENGINEERING INSPECTION REPORT

HEAD SHEAVE REPORT

Inspection & Specification Services cc.

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MECHANICAL ASSET CONDITION ASSESSMENT

Head Sheave Inspection Report

Inspection report · maintenance recommendations



Client	South32	Mine / site	Wessels Mine
Area / system	Main Shaft	Asset / asset group	Sheave Wheels
Report number	IRS32-Wessels-160726	Revision	1.2
Issue date	02/08/2026	Inspection date	16 July 2026
Issue status	Approved	Prepared by	Inspection & Specification Services cc
Reviewed by	Kevin Wood	Approved by	Kevin Wood
Client reference	South32 - Wessels	ISS job number	ISS-South32-0726-1

CONFIDENTIALITY & RELIANCE

Prepared for the South32 Wessels Mine and stated purpose only. Reliance, distribution and operational decisions remain subject to the agreed scope, requirements, stated limitations and responsible engineering review.

Controlled Document Notice

The report findings are limited to the agreed scope, access conditions, inspection methods, information supplied and evidence recorded. Visual inspection does not establish the internal condition of bearings, shafts, welds or other inaccessible components unless supporting examination or test evidence is recorded.

No external standard, acceptance criterion, measurement or NDT result may be included unless it has been verified for the specific inspection conducted.

Document Control

Table 1 - Document control and report metadata

Report title	Head Sheave Inspection Report
Report number	IRS32-Wessels-160726
Revision	1.2
Issue status	Draft
Issue date	02/08/2026
Client	South32
Mine / site	Wessels Mine
Area / system	Main Shaft
Asset / asset group	Sheave Wheels
Template version	2.1

Document Authorization Record

Table 2 - Document authorization record

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Reviewed by	Kevin Wood	Managing Engineer	iss@inspecserv.co.za	31/07/2026	KC
Approved by	Kevin Wood	Managing Engineer	iss@inspecserv.co.za	31/07/2026	KC

Revision History

Table 3 - Revision history

Revision	Date	Issue status	Description of change	Prepared by	Approved by
1.0	31/07/2026	Draft	Initial report-completion	Inspection & Specification Services cc.	Brendan Wood
1.2	07/08/2026	Draft	Findings populated from inspection notes and supplied photographic evidence.	Inspection & Specification Services cc.	Kevin Wood

Distribution List

Table 4 - Controlled distribution list

Recipient	Role	Organization	Email	Format	Distribution date
Lebohang Motete	Specialist Condition Monitoring	South32 – Wessels	Lebo.Motete@south32.net	PDF	11/08/2026
Alfonso Damon	Planner – Maintenance workforce management planning	South32 – Wessels	Alfonso.Damon@south32.net	PDF	11/08/2026
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Kevin Wood	ISS file copy	Inspection & Specification Services cc.	iss@inspecserv.co.za	PDF	11/08/2026

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SECTION A

REPORT BASIS & GOVERNANCE

Purpose, executive reporting, scope boundaries, references and responsibility interfaces.



1 Purpose

This report documents the inspection of the Underlay Sheave, Overlay Sheave and Service Winder Sheave at the South32 Wessels Mine Main Shaft. It records the agreed visual inspection scope, limitations, observations, engineering assessment framework, defect classification and maintenance recommendations arising from the inspection completed on 16 July 2026.

2 Executive Summary

The Underlay Sheave, Overlay Sheave and Service Winder Sheave were visually inspected, and their rope-groove profiles were checked using a profile. The inspection notes record approximately ~1 mm of groove wear for each sheave. The two main sheaves have a reported sheave-diameter-to-rope-diameter ratio of approximately 99:1, while the Service Winder Sheave has a reported ratio of approximately 79.5:1. Based on the screening criteria stated in the inspection notes, the recorded groove wear remains below the indicative maintenance threshold. The available evidence therefore indicates that the groove profiles continue to provide adequate support and clearance for the respective wire ropes.

The findings noted within the details of this report will form the baseline for future inspection results, as defined in the report recommendations, and summarized in Table 6 - Executive recommendation register, which defines the recommended inspection interval for as six monthly intervals, as referenced (P3 – Planned) in Table 22 - Maintenance priority guidance. Based on future scheduled inspections and findings will form the compass for possible groove wear trend and pattern corrective actions if required.

2.1 Executive Dashboard

ASSETS INSPECTED	OVERALL CONDITION	HIGHEST PRIORITY
3	Good - Minor	P4 - monitor

Table 5 - Executive asset summary matrix

Asset	Asset tag / ID	Overall condition	Highest severity	Priority	Open findings	Executive comment
Underlay Sheave	Client designation	Good	Level 1 – (Minor)	P4 - Monitor	1	~1 mm wear recorded
Overlay Sheave	Client designation	Good	Level 1 – (Minor)	P4 - Monitor	1	~1 mm wear recorded
Service Winder Sheave	Client designation	Good	Level 1 – (Minor)	P4 - Monitor	1	~1 mm wear recorded.

Table 6 - Executive recommendation register

Action ID	Asset	Recommended action	Priority	Responsible discipline	Target / shutdown	Status
REC-001	Underlay Sheave	Repeat and document profile measurement; retain nameplate and gauge evidence.	P4 - monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Open
REC-002	Overlay Sheave	Repeat and document profile measurement; retain nameplate and gauge evidence.	P4 – monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Open
REC-003	Service Winder Sheave	Repeat and document profile measurement; retain nameplate and gauge evidence.	P4 – monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Open
REC-004	All three sheaves	Obtain the relevant approved OEM/client groove-wear acceptance criterion and formalize it in the inspection procedure.	P3 - Planned	Responsible engineer / asset owner	Six Monthly Scheduled inspection from last inspection.	Open

3 Introduction

Inspection & Specification Services cc. was requested to inspect and document the condition of the Main Shaft Underlay Sheave, Overlay Sheave and Service Winder Sheave at South32 Wessels Mine. This report documents the field notes and the supplied photographic evidence for engineering review, maintenance planning and action tracking.

The head sheaves guide the respective winding ropes through the headgear arrangement. The inspection was completed on 16 July 2026 during the available access window.

4 Scope of Inspection

Table 7 - Assets and activities included in scope

Asset	Asset tag / ID	Location	Inspection activity	Access / operating state	Comment
Underlay Sheave	Client designation	Main Shaft headgear	Visual examination, photographic record and rope-groove profile check	Stopped / isolated	Photos UL-001 to UL-003 and field notes reviewed.
Overlay Sheave	Client designation	Main Shaft headgear	Visual examination, photographic record and rope-groove profile check	Stopped / isolated	Photos OL-001 to OL-002 and field notes reviewed.
Service Winder Sheave	Client designation	Main Shaft headgear	Visual examination, photographic record and rope-groove profile check	Stopped / isolated	Photos SW-001 to SW-003 and field notes reviewed.

Table 8 - Scope exclusions

Excluded item	Reason / access constraint	Assessment implication	Recommended follow-up
Internal bearing inspection	No bearing disassembly or internal examination recorded, not part of the inspection scope.	As part of this inspection internal bearing condition cannot be confirmed.	Review internal maintenance records and perform specialist inspection where required.
Dynamic load / operational test	No operational test data included not part of the inspection scope.	Dynamic behavior and loaded tracking are not assessed.	Obtain operating observations, vibration and temperature data if required as per mine defined condition monitoring schedule.
NDT and formal alignment survey	No verified NDT report or alignment readings supplied for this exercise, not part of the inspection scope.	Crack condition and alignment tolerances are not established.	Perform by other independent third party appointed by the mine.

5 Inspection Limitations

The inspection was limited by the recorded access, operating condition, visibility, cleanliness, disassembly status, information supplied and inspection methods. Internal bearing condition must not be inferred from external visual inspection unless supporting evidence is available.

NDT results may be reported only where the method, scope, operator or report reference and result evidence are available.

Alignment, bearings and internal components

Table 9 - Inspection limitations register

Limitation ID	Affected asset / area	Limitation	Impact on assessment	Follow-up requirement
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6 Applicable Standards and References

Table 10 - Applicable standards and references register

Ref. ID	Type	Title / description	Document number	Revision / date	Source	Status
REF-001	OEM nameplate	Underlay Sheave identification and groove-data plates	Order 4541088384 / Job 4918	Manufactured 05-2018	PHOTO-UL-002	Reference evidence
REF-002	OEM nameplate	Service Winder Sheave identification plate	Order 4200391544 / Job 0474	Manufactured 03-2005	PHOTO-SW-002	Reference evidence
REF-003	Field notes	Main Head Sheave Notes - geometric and wear calculations	Not assigned	Provided 2026	ISS	Preliminary source;
REF-004	Photographic record	Underlay, Overlay and Service Winder inspection image set	PHOTO-UL/OL/SW series	16/07/2026	ISS	Included
REF-005	Reference schematic	Main Shaft head sheave orientation schematic	Figure 1	2026	ISS	Reference only; not to scale

7 Definitions and Abbreviations

Table 11 - Definitions and abbreviations

Term	Definition
IR	Inspection Report
ISS	Inspection & Specification Services cc.
NDT	Non-destructive testing
UT	Ultrasonic testing
MPI	Magnetic particle inspection
OEM	Original equipment manufacturer
P1-P4	Maintenance priority classifications defined in this report

8 Responsibilities

Table 12 - Responsibility and interface matrix

Party / role	Responsibility	Report interface / deliverable
Client management / responsible site authority	Provide scope confirmation, access, isolation, asset information, operational context and formal site decisions.	Review findings, determine operational response and authorize corrective actions.
ISS inspector / project manager	Perform the agreed inspection, record evidence, compile findings and recommendations within the stated limitations.	Issue draft and final inspection report.
ISS reviewing engineer	Review technical content, evidence traceability, engineering assessment and recommendations.	Technical review and approval.
Maintenance contractor / specialist, where appointed	Verify repair scope, execute approved work and provide completion evidence.	Repair records, measurements, NDT and close-out evidence.

SECTION B

ASSET & INSPECTION FRAMEWORK

Asset data, inspection equipment, methodology, assessment criteria and reference orientation.



9 Asset Information

9.1 Underlay Sheave Asset Data Sheet

Table 13 - Underlay Sheave Asset Data Sheet - asset data sheet

Field	Entry	Field	Entry
Asset name	Underlay Sheave	Asset tag / ID	Client designation - Underlay
Location	Wessels Mine Main Shaft headgear	Service / duty	Main shaft head sheave - underlay position
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Order 4541088384; Job 4918
Sheave diameter	3658 mm / 12 ft (nameplate)	Number of grooves	Single groove
Rope diameter	37 mm (nameplate)	Recorded groove wear	~1 mm (field notes; measurement record)
Diameter ratio D:d	99:1 (calculated)	Groove Wear to rope ratio	Current wear 2.70% (calculated)
Mass	4140 kg (nameplate)	Moment of inertia	6575 kg·m ² as marked
Date manufactured	05-2018 (nameplate)	OEM groove data	A = 117 mm; B = 142 mm; 22 min marking
Inspection date	16 July 2026	Overall condition rating	Good

9.2 Overlay Sheave Asset Data Sheet

Table 14 - Overlay Sheave Asset Data Sheet - asset data sheet

Field	Entry	Field	Entry
Asset name	Overlay Sheave	Asset tag / ID	Client designation - Overlay
Location	Wessels Mine Main Shaft headgear	Service / duty	Main shaft head sheave - overlay position
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd. (field notes)	Model / serial number	Not supplied / Inaccessible
Sheave diameter	3658 mm (field notes)	Number of grooves	Single groove
Rope diameter	37 mm (field notes)	Recorded groove wear	~1 mm (field notes)
Diameter ratio D:d	98.9:1 (calculated)	Wear to rope ratio	2.70% (calculated)
Inspection date	16 July 2026	Overall condition rating	Good
Reference	PHOTO-OL-001 and PHOTO-OL-002		

9.3 Service Winder Sheave Asset Data Sheet

Table 15 - Service Winder Sheave Asset Data Sheet - asset data sheet

Field	Entry	Field	Entry
Asset name	Service Winder Sheave	Asset tag / ID	Client designation - Service Winder
Location	Wessels Mine Main Shaft headgear	Service / duty	Service winder rope sheave
Manufacturer / OEM	Kimberley Engineering Works (Pty) Ltd.	Model / serial number	Order 4200391544; Job 0474
Sheave diameter	1828 mm (nameplate)	Number of grooves	Single groove
Rope diameter	23 mm (nameplate)	Recorded groove wear	~1 mm (field notes; verify measurement record)
Diameter ratio D:d	79.5:1 (calculated)	Wear to rope ratio	4.35% (calculated)
Mass	1500 kg (nameplate)	Moment of inertia	675 kg·m²
Date manufactured	03-2005 (nameplate)	Mine / shaft marking	Samancor / Wessels Mine
Inspection date	16 July 2026	Overall condition rating	Good

10 Inspection Team and Equipment Used

10.1 Inspection Team

Table 16 - Inspection team register

Name	Role / discipline	Company	Competency / authorization	Site induction / permit reference
Brendan Wood	Inspector / Projects and Systems	ISS	Technician	N/A
Kevin Wood	Technical reviewer / Managing Engineer	ISS	Managing Engineer	N/A

10.2 Inspection Equipment

Table 17 - Inspection equipment and calibration register

Equipment	Purpose	Manufacturer / model	Equipment ID	Calibration certificate	Calibration due	Accuracy / range	Comment
Flexicurve profile	Comparative profile / wear check	Comparative profile / wear check	Not recorded	N/A	N/A	N/A	Visible in PHOTO-UL-001, PHOTO-OL-001 and PHOTO-SW-001.
Digital camera / mobile device	Photographic evidence records	Not recorded	Not recorded	N/A	N/A	Image record	File timestamps used for traceability.

11 Inspection Methodology

The inspection methodology may combine visual examination, photographic documentation, dimensional measurement, rope groove assessment, rope tracking and alignment observation.

Table 18 - Inspection methodology matrix

Method ID	Inspection activity	Purpose	Evidence captured	Acceptance / reference basis
M-01	Visual inspection	Record visible condition, wear, contamination and anomalies	Field notes and photographs	Engineering assessment within stated limitations
M-02	Rope groove profile check	Record comparative groove profile and wear	Flexicurve-gauge photographs and field-note values	OEM/client criterion to be verified
M-03	Asset identification	Capture OEM/nameplate information	PHOTO-UL-002 and PHOTO-SW-002	Nameplate / client asset register
M-04	Rope tracking / alignment observation	Identify visible side-contact or approach/departure concerns	Limited contextual photographs	Verified by observed wear pattern

12 Inspection Criteria

Table 19 - Inspection criteria and evidence requirements

Attribute	Assessment basis	Evidence required	Typical record	Reference / note
Rope groove and rim condition	Visible wear, damage, profile concerns and abnormal contact evidence	Photographs and measurements were performed	Finding / defect ID	[Reference]
Rope tracking	Visible tracking evidence, contact pattern	Photographs, alignment observations	Finding / defect ID	[Reference]

13 Condition Rating, Defect Severity and Maintenance Priority

The following classifications support consistent reporting and maintenance planning. They do not replace the mine's formal risk assessment, statutory obligations or decisions by the responsible site authority.

Table 20 - Condition rating

Rating	General meaning	Typical reporting response
Excellent	No material defect observed within the inspection scope.	Record condition and routine inspection interval.
Good	Minor condition observations only.	Monitor through routine maintenance and inspection.
Fair	Serviceable with deterioration requiring planned attention.	Plan maintenance, measurement or enhanced monitoring.
Poor	Significant deterioration requiring priority engineering or maintenance review.	Define corrective action and verification requirements.
Critical	Potentially unacceptable condition requiring immediate responsible-site review.	Escalate immediately and determine operational response.

Table 21 - Defect severity classification

Level	Classification	Definition	Default action
Level 1	Minor	Low-significance observation requiring monitoring or routine correction.	Record and monitor.
Level 2	Moderate	Defect requiring planned corrective action.	Schedule corrective work.
Level 3	Significant	Defect requiring priority engineering review and corrective action.	Repair at the next suitable shutdown or earlier.
Level 4	Critical	Potentially critical condition requiring immediate responsible-site review.	Immediate escalation and corrective action.

Table 22 - Maintenance priority guidance

Priority	Meaning	Typical timeframe guidance	Required response
P1	Immediate	Before continued operation where required by responsible review.	Escalate and define immediate controls / corrective action.
P2	Urgent	Earliest practicable maintenance opportunity.	Plan and execute short-term corrective action.
P3	Planned	Next suitable scheduled shutdown or maintenance cycle.	Include in approved maintenance plan.
P4	Monitor	Routine inspection and maintenance interval.	Trend and verify at future inspection.

14 Engineering/Reference Sketches

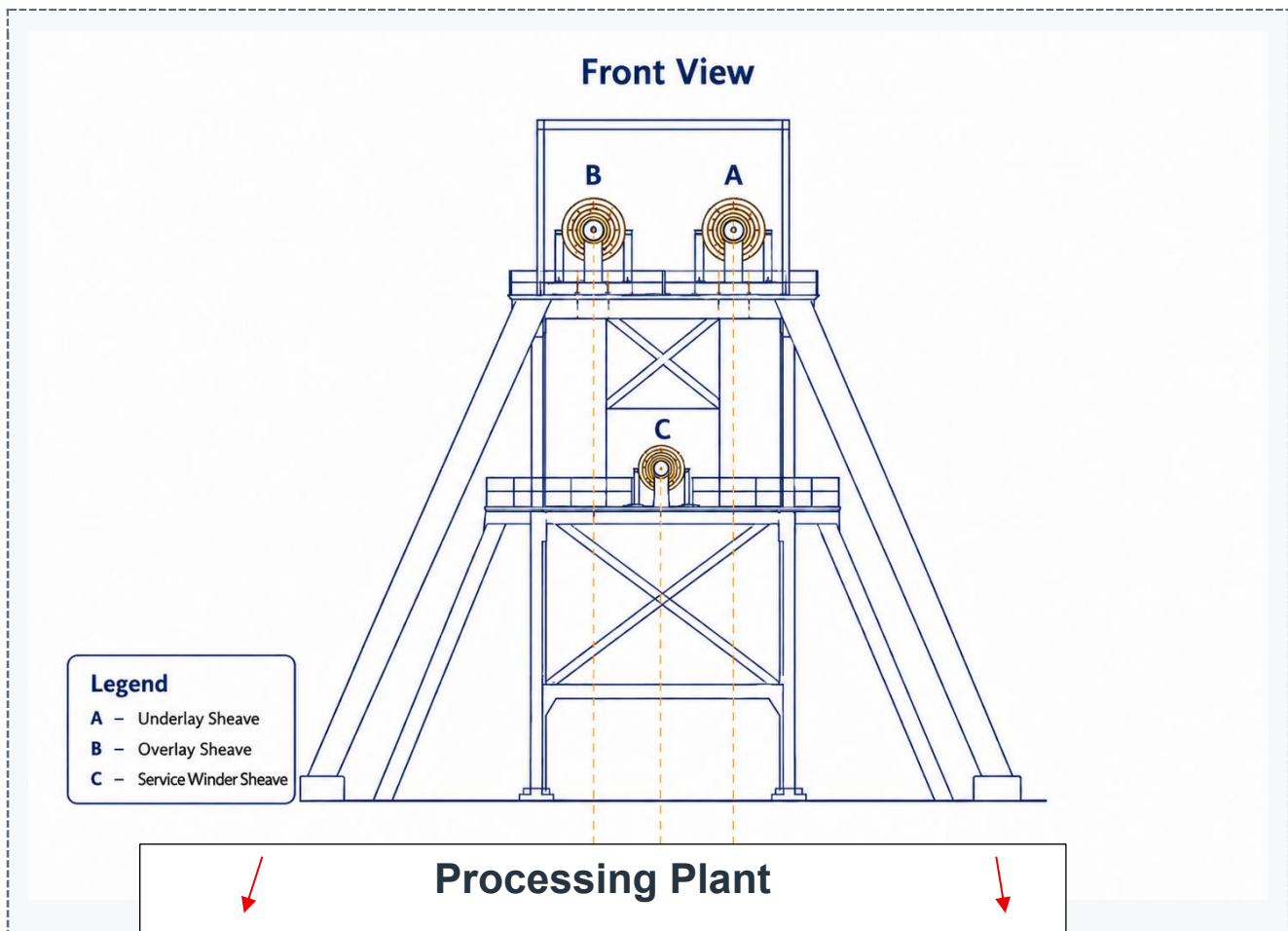


Figure 1 – Reference arrangement of the main shaft head sheaves (view schematic, not to scale)

Table 23 - Inspection orientation key

Reference	Description	Direction / datum	Source drawing / photograph
Overlay Sheave	Positional reference location A	Front View	Figure 1
Underlay Sheave	Positional reference location B	Front View	Figure 1
Service Winder Sheave	Positional reference location C	Front View	Figure 1

SECTION C**FINDINGS, RISK & ACTIONS**

Repeatable asset findings, defect traceability, engineering assessment and maintenance recommendations.



15 Asset Descriptions

15.1 Underlay Sheave

The Underlay Sheave is positioned within the Wessels Mine Main Shaft headgear and guides the corresponding main winding rope through the underlay rope path. The OEM plates identify a 3658 mm single-groove sheave designed for a 37 mm rope. The field notes record ~1 mm of groove wear. The supplied close-up photographs principally document the groove profile, Lubrication and the OEM identification data.

Table 24 - Underlay Sheave - asset description summary

Field	Entry
Asset	Underlay Sheave
Operational duty	Main shaft winding rope - underlay path
Location	Wessels Mine Main Shaft headgear
Principal components	Sheave rim/groove, hub, shaft/axle, bearings, supports, guards and fasteners
Reference	PHOTO-UL-001 to PHOTO-UL-003; OEM Order 4541088384 / Job 4918
Inspection access state	Stopped / isolated state

15.2 Overlay Sheave

The Overlay Sheave is the companion main shaft head sheave serving the overlay rope path. The field notes identify a 3658 mm sheave operating with a 37 mm rope and record ~1 mm of groove wear. The supplied photographs document the groove-profile gauge position and the visible groove surface; the OEM identification plate was not included in the supplied Overlay set as the plate was inaccessible due to sheave orientation at the time of inspection.

Table 25 - Overlay Sheave - asset description summary

Field	Entry
Asset	Overlay Sheave
Operational duty	Main shaft winding rope - overlay path
Location	Wessels Mine Main Shaft headgear
Principal components	Sheave rim/groove, hub, shaft/axle, bearings, supports, guards and fasteners
Reference	PHOTO-OL-001 and PHOTO-OL-002; field notes
Inspection access state	Stopped / isolated state to be confirmed

15.3 Service Winder Sheave

The Service Winder Sheave guides the service-winder rope at the Main Shaft. The supplied nameplate identifies a 1828 mm sheave designed for a 23 mm rope, with a mass of 1500 kg. The field notes record ~1 mm of groove wear. The supplied images document the groove-profile gauge position, the nameplate and the general groove surface.

Table 26 - Service Winder Sheave - asset description summary

Field	Entry
Asset	Service Winder Sheave
Operational duty	Service winder rope guidance
Location	Wessels Mine Main Shaft headgear
Principal components	Sheave rim/groove, hub, shaft/axle, bearings, supports, guards and fasteners
Reference	PHOTO-SW-001 to PHOTO-SW-003; OEM Order 4200391544 / Job 0474
Inspection access state	Stopped / isolated state

16 Inspection Findings

16.1 Underlay Sheave

16.1.1 Asset Overview and General Condition

Underlay Sheave | Main Shaft headgear | Condition Rating: Good - Minor condition observations only.

The field notes record ~1 mm of groove wear on a 3658 mm sheave operating with a 37 mm rope. The supplied photographs show the profile positioned in the groove and the OEM identification plates.

16.1.2 Inspection Observation

PHOTO-UL-001 shows the profile positioned against the groove profile. The field notes record ~1 mm wear.
PHOTO-UL-002 records the OEM nameplate information.
PHOTO-UL-003 shows a broader view of the groove surface.

16.1.3 Rope Groove and Rim Condition

The recorded ~1 mm groove wear represents approximately 2.70% of the 37 mm nominal rope diameter. The diameter-to-rope ratio is approximately 99:1. The notes compare the value with an indicative 3.7 mm to 4.6 mm screening range.

16.1.4 Rope Tracking and Alignment

Rope tracking: wear pattern observed shows normal and properly aligned within the groove.

Rim condition: uniform wear with no signs of side-loading or improper contact.

Overall status: Good | Monitor | No remedial actions required.

16.1.5 Corrosion, Coating and Housekeeping

Corrosion status: No signs of surface rust, pitting, or structural corrosion found.

Coating condition: Paint coating is fully intact, undamaged, and provides adequate protection.

Lubrication state: Operating groove is evenly and properly coated with grease across the entire groove surface.

16.1.6 Engineering Assessment

ENGINEERING INTERPRETATION

The recorded wear ~1mm wear is small relative to the nominal rope diameter and below the indicative threshold stated in the notes. The groove profile still provides adequate clearance and support for a 37 mm steel rope without risk of crushing or pinching the strands.

16.1.7 Defect Classification and Recommended Action

Table 27 - DEF-001 Underlay sheave groove finding classification and action

Finding ID	Condition rating	Severity	Priority	Future Recommended action	Responsible discipline	Verification requirement
DEF-001	Good	Level 1	P4 - monitor	Clean the groove, identify the gauge, repeat and record readings at defined circumferential positions, and compare the results with the approved OEM/client criterion.	Mechanical maintenance / competent inspector	Post-cleaning photographs, gauge used to reference profile, measurement sheet and approved criterion comparison

16.1.8 Photographic Evidence Sheet

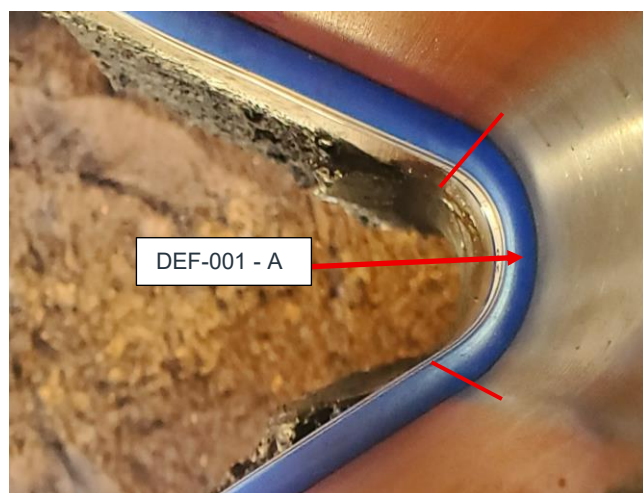


Photo ID
PHOTO-UL-001

Finding ID
DEF-001

Asset
Underlay Sheave

Component / location
Rope groove / rim

View / direction
Close-up of groove profile

Date / time
16/07/2026 11:43 (filename timestamp)

Caption

Profile gauge positioned in the Underlay Sheave groove. The field notes record ~1 mm wear.

Figure 2 - Underlay Sheave evidence - PHOTO-UL-001 / DEF-001

Table 28 - Photographic evidence assessment - PHOTO-UL-001

Observation	Engineering assessment	Recommended action
Profile is visible against the groove profile; the notes record ~1 mm wear at position DEF-001-A.	The value is within acceptable wear limits The value of ~1 mm of wear, the profile sits within its normal operating limits and is below the replacement or repair thresholds. The groove profile still provides adequate clearance and support for a 37 mm steel rope without risk of crushing or pinching the strands.	Clean, repeat and document the measurement using an identified gauge.

16.2 Overlay Sheave

16.2.1 Asset Overview and General Condition

Overlay Sheave | Main Shaft headgear | Condition Rating: Good - Minor condition observations only.

The field notes identify a 3658 mm sheave operating with a 37 mm rope and record ~1 mm of groove wear. The supplied photographs show the profile gauge and the visible groove surface.

16.2.2 Inspection Observation

PHOTO-OL-001 shows the profile positioned against the groove profile. The field notes record ~1 mm wear.
PHOTO-OL-002 shows a broader view of the groove surface.

16.2.3 Rope Groove and Rim Condition

The recorded ~1 mm wear represents approximately 2.70% of the 37 mm nominal rope diameter. The reported diameter-to-rope ratio is approximately 99:1. The notes compare the value with an indicative 3.7 mm to 4.6 mm screening range or upper limits.

16.2.4 Rope Tracking and Alignment

Rope tracking: wear pattern observed shows normal and properly aligned within the groove.

Rim condition: Uniform wear with no signs of side-loading or improper contact.

Overall status: Good | Monitor | No remedial actions required.

16.2.5 Corrosion, Coating and Housekeeping

Corrosion status: No signs of surface rust, pitting, or structural corrosion found.

Coating condition: Paint coating is fully intact, undamaged, and provides adequate protection.

Lubrication state: Operating groove is evenly and properly coated with grease across the entire groove surface.

16.2.6 Engineering Assessment

ENGINEERING INTERPRETATION

The recorded wear ~1mm wear is small relative to the nominal rope diameter and below the indicative threshold stated in the notes. The groove profile still provides adequate clearance and support for a 37 mm steel rope without risk of crushing or pinching the strands.

16.2.7 Defect Classification and Recommended Action

Table 29 - DEF-002 Overlay sheave finding classification and action

Finding ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-002	Good	Level 1	P4 - monitor	Clean the groove, repeat the profile check at defined positions, record the gauge and readings, and compare against the approved criterion.	Mechanical maintenance / competent inspector	Post-cleaning photographs, gauge ID, measurement sheet and approved criterion comparison

16.2.8 Photographic Evidence Sheet



Photo ID
PHOTO-OL-001

Finding ID
DEF-002

Asset
Overlay Sheave

Component / location
Rope groove / rim

View / direction
Close-up of groove profile

Date / time
16/07/2026 11:51 (filename timestamp)

Caption
Profile positioned in the Overlay Sheave groove. The field notes record ~1 mm wear.

Figure 3 - Overlay Sheave evidence - PHOTO-OL-001 / DEF-002

Table 30 - Photographic evidence assessment - PHOTO-OL-001

Observation	Engineering assessment	Recommended action
Profile is visible against the groove profile; the notes record ~1 mm wear at position DEF-002-A.	The value is within acceptable wear limits The value of ~1 mm of wear, the profile sits within its normal operating limits and is below the replacement or repair thresholds. The groove profile still provides adequate clearance and support for a 37 mm steel rope without risk of crushing or pinching the strands.	Clean, repeat and document the measurement using an identified gauge.

16.3 Service Winder Sheave

16.3.1 Asset Overview and General Condition

Service Winder Sheave | Main Shaft headgear | Preliminary condition: Good.

The OEM nameplate identifies a 1828 mm sheave operating with a 23 mm rope. The field notes record ~1 mm of groove wear. The supplied photographs show the profile gauge and the visible groove surface.

16.3.2 Inspection Observation

PHOTO-SW-001 shows the profile positioned against the groove profile. The field notes record ~1 mm wear.
PHOTO-SW-002 records the OEM nameplate information.
PHOTO-SW-003 shows a broader view of the groove surface

16.3.3 Rope Groove and Rim Condition

The recorded ~1 mm wear represents approximately 4.35% of the 23 mm nominal rope diameter. The diameter-to-rope ratio is approximately 79.5:1. The notes compare the value with an indicative 2.3 mm to 2.9 mm screening range or upper limits.

16.3.4 Rope Tracking and Alignment

Rope tracking: wear pattern observed shows normal and properly aligned within the groove.

Rim condition: uniform wear with no signs of side-loading or improper contact.

Overall status: Good | Monitor | No remedial actions required.

16.3.5 Corrosion, Coating and Housekeeping

Corrosion status: No signs of surface rust, pitting, or structural corrosion found.

Coating condition: Paint coating is fully intact, undamaged, and provides adequate protection.

Lubrication state: Operating groove is evenly and properly coated with grease across the entire groove surface.

16.3.6 Engineering Assessment

ENGINEERING INTERPRETATION

The recorded wear ~1mm wear is small relative to the nominal rope diameter and below the indicative threshold stated in the notes. The groove profile still provides adequate clearance and support for a 23 mm steel rope without risk of crushing or pinching the strands.

16.3.7 Defect Classification and Recommended Action

Table 31 - DEF-003 Service Winder finding classification and action

Finding ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-003	Good	Level 1	P4 - monitor	Repeat and document the profile measurement at defined positions, record the gauge ID, and verify the result against the approved OEM/client criterion.	Mechanical maintenance / competent inspector	Gauge ID, measurement sheet, approved criterion comparison and follow-up photographs

16.3.8 Photographic Evidence Sheet

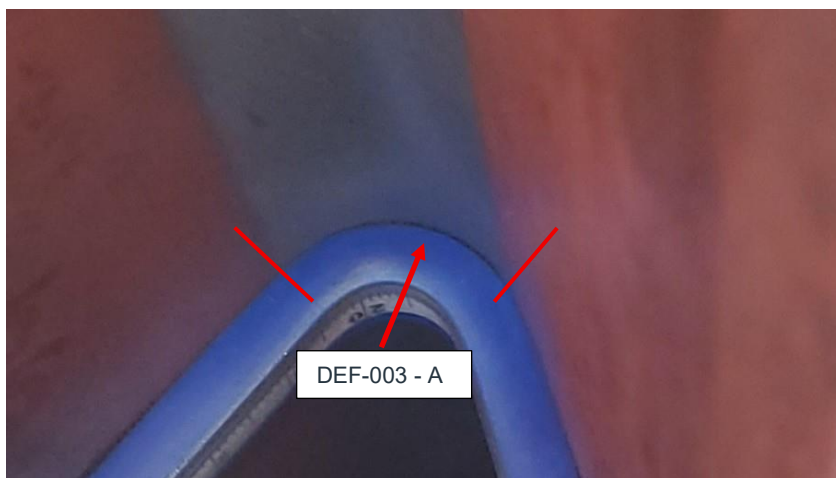


Photo ID
PHOTO-SW-001

Finding ID
DEF-003

Asset
Service Winder Sheave

Component / location
Rope groove / rim

View / direction
Close-up of groove profile

Date / time
16/07/2026 12:12 (filename timestamp)

Caption
Profile gauge positioned in the Service Winder Sheave groove. The field notes record ~1 mm wear.

Figure 4 - Service Winder Sheave evidence - PHOTO-SW-001 / DEF-003

Table 32 - Photographic evidence assessment - PHOTO-SW-001

Observation	Engineering assessment	Recommended action
Profile is visible against the groove profile; the notes record ~1 mm wear at position DEF-003-A.	The value is within acceptable wear limits The value of ~1 mm of wear, the profile sits within its normal operating limits and is below the replacement or repair thresholds. The groove profile still provides adequate clearance and support for a 23 mm steel rope without risk of crushing or pinching the strands.	Clean, repeat and document the measurement using an identified gauge

17 Summary of Defects

Table 33 - Consolidated defect register

Finding ID	Asset	Component / location	Finding / observation	Condition	Severity	Priority	Evidence	Responsible discipline	Status
DEF-001	Underlay Sheave	Rope groove	1 mm wear recorded;	Good	Level 1 – (Minor)	P4 - Monitor	PHOTO-UL-001 to 003	Mechanical maintenance / inspection	Open
DEF-002	Overlay Sheave	Rope groove	1 mm wear recorded; measurement	Good	Level 1 – (Minor)	P4 – Monitor	PHOTO-OL-001 to 002	Mechanical maintenance / inspection	Open
DEF-003	Service Winder Sheave	Rope groove	1 mm wear recorded;	Good	Level 1 – (Minor)	P4 - Monitor	PHOTO-SW-001 to 003	Mechanical maintenance / inspection	Open

18 Inspection Observation Log

Table 34 - Inspection observation and evidence log

Observation ID	Asset	Location	Observation	Evidence reference	Condition rating	Related finding ID	Status
OBS-001	Underlay Sheave	Rope groove	Profile shown; field notes record 1 mm wear	PHOTO-UL-001	Good	DEF-001	Recorded
OBS-002	Underlay Sheave	OEM plates	OEM identification and groove design information recorded.	PHOTO-UL-002	Information only	None	Recorded
OBS-003	Underlay Sheave	Groove / adjacent surface	Contextual groove and surface view.	PHOTO-UL-003	Information only	DEF-001	Recorded
OBS-004	Overlay Sheave	Rope groove	Profile gauge shown; field notes record 1 mm wear.	PHOTO-OL-001	Good	DEF-002	Recorded
OBS-005	Overlay Sheave	Rope groove	Contextual groove surface view.	PHOTO-OL-002	Information only	DEF-002	Recorded
OBS-006	Service Winder Sheave	Rope groove	Profile gauge shown; field notes record 1 mm wear.	PHOTO-SW-001	Good	DEF-003	Recorded
OBS-007	Service Winder Sheave	OEM nameplate	1828 mm sheave and 23 mm rope data recorded.	PHOTO-SW-002	Information only	None	Recorded
OBS-008	Service Winder Sheave	Rope groove	Contextual groove surface view.	PHOTO-SW-003	Information only	DEF-003	Recorded

19 Engineering Assessment

CONSOLIDATED ASSESSMENT

The field notes record ~1 mm of groove wear on each of the three inspected sheaves. This equates to approximately 2.70% of the 37 mm rope diameter for the Underlay and Overlay Sheaves and approximately 4.35% of the 23 mm rope diameter for the Service Winder Sheave. The corresponding D:d ratios are approximately 98.9:1 and 79.5:1. On the indicative screening basis stated in the notes, the recorded wear is below the quoted intervention ranges. The measurements are therefore classified as minor and suitable for monitoring rather than immediate machining, and fit for operational use.

20 Recommendations

To formalize the trends in the wear pattern and groove profile a six-monthly interval for follow up inspections would be required to measure and capture quantitative wear rates value. as referenced (P3 – Planned) in Table 22 - Maintenance priority guidance.

Action ID	Related finding ID	Asset	Recommendation	Priority	Responsible discipline	Target / shutdown	Verification requirement	Close-out evidence	Status
REC-001	DEF-001	Underlay Sheave	Clean the groove and repeat the profile measurement at defined circumferential positions using an identified gauge.	P4 - Monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Recorded readings and approved criterion comparison	Measurement sheet and post-cleaning photos	Open
REC-002	DEF-002	Overlay Sheave	Clean as required and repeat the profile measurement at defined circumferential positions using an identified gauge.	P4 – Monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Recorded readings and approved criterion comparison	Measurement sheet and follow-up photos	Open
REC-003	DEF-003	Service Winder Sheave	Repeat and document the profile measurement at defined circumferential positions using an identified gauge.	P4 – Monitor	Mechanical maintenance / competent inspector	Next routine maintenance opportunity	Recorded readings and approved criterion comparison	Measurement sheet and follow-up photos	Open
REC-004	All findings	All three sheaves	Obtain and reference the current OEM/client sheave-groove acceptance criterion before final report approval.	P3 - Planned	Responsible engineer / asset owner	Six Monthly Scheduled inspection from last inspection.	Approved reference incorporated into report	Controlled reference / approval record	Open

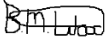
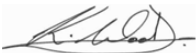
21 Conclusion

The preliminary review of the supplied notes and photographs indicates that each of the three sheaves has a recorded groove-wear value of approximately ~1 mm. Relative to the nominal rope diameters, the recorded wear is low and falls below the indicative screening ranges quoted in the notes. No immediate groove machining or replacement is indicated by the supplied information alone. The sheaves are provisionally rated Good with Level 1 findings and P4 monitoring actions. Final acceptance remains subject to a traceable repeat measurement, verification of the applicable OEM/client criterion and completion of any alignment, bearing, structural or NDT assessment required by the responsible site authority.

22 Technical Review and Report Issue Sign-off

The sign-off below confirms document review only and does not replace the client's operational risk assessment or responsibility for corrective actions.

Table 35 - Technical review and issue sign-off

Role	Name	Title / discipline	Review status	Date	Signature
Prepared by	Brendan Wood	Projects / Systems	Report preparation	02/08/2026	
Technical reviewer	Kevin Wood	Managing Engineer	Review Complete.	02/08/2026	
Approved for issue	Kevin Wood	Managing Engineer	Approval	04/08/2026	

Appendix A - Photographic Register

Table 36 - Photographic register

Photo ID	Asset	Component / location	View / direction	Description / observation	Related finding ID	Date / time	File name / source	Report location
PHOTO-UL-001	Underlay Sheave	Rope groove / rim	Close-up with profile gauge	Field notes record 1 mm wear; contamination visible.	DEF-001	16/07/2026 11:43	PHOTO-UL-001(20260716_1 14341).jpg	Section 16 / Appendix A
PHOTO-UL-002	Underlay Sheave	OEM identification plates	Side elevation	OEM identification and groove design information.	None	16/07/2026 12:15	PHOTO-UL-002(20260716_1 21536).jpg	Appendix A
PHOTO-UL-003	Underlay Sheave	Rope groove / adjacent surface	Context view	Additional groove and adjacent surface condition.	DEF-001	16/07/2026 11:56	PHOTO-UL-003(20260716_1 15626).jpg	Appendix A
PHOTO-OL-001	Overlay Sheave	Rope groove / rim	Close-up with profile gauge	Field notes record 1 mm wear.	DEF-002	16/07/2026 11:51	PHOTO-OL-001(20260716_1 15135).jpg	Section 16 / Appendix A
PHOTO-OL-002	Overlay Sheave	Rope groove / rim	Context view	Visible groove surface and surrounding structure.	DEF-002	16/07/2026 11:56	PHOTO-OL-002(20260716_1 15654).jpg	Appendix A
PHOTO-SW-001	Service Winder Sheave	Rope groove / rim	Close-up with profile gauge	Field notes record 1 mm wear.	DEF-003	16/07/2026 12:12	PHOTO-SW-001(20260716_1 21216).jpg	Section 16 / Appendix A
PHOTO-SW-002	Service Winder Sheave	OEM identification plate	Nameplate close-up	1828 mm sheave, 23 mm rope and 1500 kg mass recorded.	None	16/07/2026 12:14	PHOTO-SW-002(20260716_1 21459).jpg	Appendix A
PHOTO-SW-003	Service Winder Sheave	Rope groove / rim	Context view	Visible groove surface condition.	DEF-003	16/07/2026 12:27	PHOTO-SW-003(20260716_1 22749).jpg	Appendix A

A.1 Underlay Sheave - Supporting Photographs



PHOTO-UL-002

Kimberley Engineering Works identification and groove-data plates used to populate the asset data sheet.



PHOTO-UL-003

Contextual view of the Underlay Sheave groove and adjacent surface.

Figure 5 - Underlay Sheave supporting evidence - PHOTO-UL-002 and PHOTO-UL-003

A.2 Overlay Sheave - Supporting Photographs

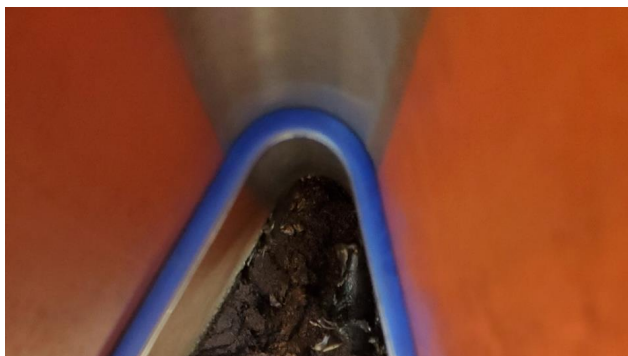


PHOTO-OL-001

Profile gauge positioned in the groove; the field notes record 1 mm wear.

PHOTO-OL-002

Contextual view of the Overlay Sheave groove surface and surrounding structure.

Figure 6 - Overlay Sheave supporting evidence - PHOTO-OL-001 and PHOTO-OL-002

A.3 Service Winder Sheave - Supporting Photographs

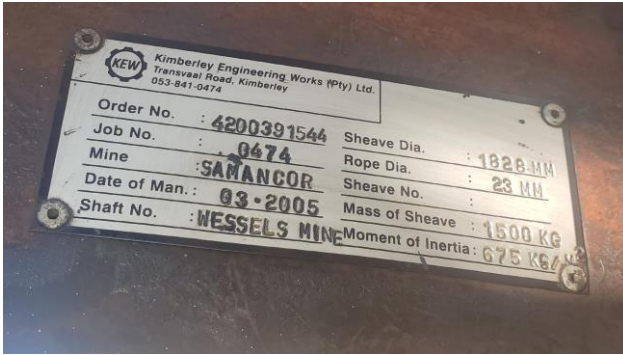


PHOTO-SW-002
Kimberley Engineering Works nameplate recording 1828 mm sheave diameter, 23 mm rope diameter and 1500 kg mass.

PHOTO-SW-003
Contextual view of the Service Winder Sheave groove surface.

Figure 7 - Service Winder Sheave supporting evidence - PHOTO-SW-002 and PHOTO-SW-003

Appendix B - Inspection Checklist

Table 37 - Head sheave inspection checklist

Asset	Inspection check	Result	Observation / limitation	Evidence reference	Related finding ID
Underlay Sheave	Asset identification confirmed	Pass	Identification evidence available;	PHOTO-UL-002	None
Underlay Sheave	Rope groove and rim condition examined	Pass	Profile gauge photographed and 1 mm wear recorded in notes.	PHOTO-UL-001	DEF-001
Overlay Sheave	Asset identification confirmed	Requires Confirmation	Identification evidence available; Overlay nameplate inaccessible.	Field notes	None
Overlay Sheave	Rope groove and rim condition examined	Pass	Profile photographed and 1 mm wear recorded in notes.	PHOTO-OL-001	DEF-002
Service Winder Sheave	Asset identification confirmed	Pass / partial	Identification evidence available.	PHOTO-SW-002	None
Service Winder Sheave	Rope groove and rim condition examined	Pass	Profile photographed and 1 mm wear recorded in notes.	PHOTO-SW-001	DEF-003

Appendix C - Measurements and Dimensional Records

Table 38 - Measurements and dimensional records

Measurement ID	Asset	Location / feature	Instrument / equipment ID	Measured value	Unit	Reference / criterion	Calculated comparison	Comment
MEAS-UL-001	Underlay Sheave	Rope groove profile	Profile gauge - ID not recorded	1	mm	Approved OEM/client criterion outstanding	2.7% of 37 mm rope diameter	Field-note value; verify.
MEAS-OL-001	Overlay Sheave	Rope groove profile	Profile gauge - ID not recorded	1	mm	Approved OEM/client criterion outstanding	2.7% of 37 mm rope diameter	Field-note value; verify.
MEAS-SW-001	Service Winder Sheave	Rope groove profile	Profile gauge - ID not recorded	1	mm	Approved OEM/client criterion outstanding	4.35% of 23 mm rope diameter	Field-note value; verify.

Appendix D - Drawings, Schematics and Reference Information

Table 39 - Drawings and reference information register

Reference ID	Drawing / document number	Title	Revision	Source	Status	Comment
SK-001	Figure 1	Reference arrangement of the main shaft head sheaves	2026	ISS	Reference only	Schematic, not to scale.
OEM-UL-001	Order 4541088384 / Job 4918	KEW Underlay Sheave nameplate and groove plate	Manufactured 05-2018	PHOTO-UL-002	Reference evidence	Verify against client asset register.
OEM-SW-001	Order 4200391544 / Job 0474	KEW Service Winder Sheave nameplate	Manufactured 03-2005	PHOTO-SW-002	Reference evidence	Verify against client asset register.

Appendix F - Supporting Documentation

Table 40 - Supporting documentation register

Document ID	Title / description	Document number	Revision / date	Source	Report relevance	Status
DOC-001	Underlay Sheave photographic record	PHOTO-UL series	16/07/2026	ISS	Supports groove and asset identification observations	Included
DOC-002	Overlay Sheave photographic record	PHOTO-OL series	16/07/2026	ISS	Supports groove observations	Included
DOC-003	Service Winder Sheave photographic record	PHOTO-SW series	16/07/2026	ISS	Supports groove and asset identification observations	Included